

Intro 1: Visual Perception Revision Lecture

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Quiz-based revision – why?

- **What's the most successful revision strategy?**
 - **Self-testing** is the best predictor for a good exam mark!

Gurung , R.A.R. (2005). How do students really study (and does it matter)?
Teaching of Psychology, 32(4), 238-240.
 - Answer the items on the **learning goal slides**, the **MCQs** for each lecture and the formative class test,
 - ...and your **own questions**.
 - Test yourself and test/be tested by your colleagues.

Lecture 1: Basics of vision I - the eye

- **Complete the sentences from a choice of options.**

The process with which our eyes focus on an object is referred to as _ _ _ .

- | | |
|-----------------|------------------|
| 1) assimilation | 2) accommodation |
| 3) bundling | 4) conversion |

Two parts of the eye are chiefly involved in this process: the _ _ _ and the _ _ _ . The result of this process is that the object at the focus point is represented in the _ _ _ .

- | | | |
|--------------|-------------|-----------------------|
| a) ...retina | b) ...fovea | c) ...ciliary muscles |
| d) ...cornea | e) ...lens | f) ...optic disc |

Lecture 1: Basics of vision I - the eye

- **How does accommodation work?**

Complete the sentences.

To accommodate to **near objects**, the ciliary muscles are _ _ _ _
so that the lens is _ _ _ _ _
so that the light waves are bent _ _ _ _ _ .

To accommodate to **distant objects**, the ciliary muscles are _ _ _ _
so that the lens is _ _ _ _ _
so that the light waves are bent _ _ _ _ _ .

Lecture 1: Basics of vision I - the eye

- **Open question: Visual sensation includes two types of transformations. What are they and what happens there?**

Lecture 1: Basics of vision I - the eye

- **Which alternative leads to a correct sentence?**
 - a) There are (more/less) cones than rods in the retina.
 - b) In the fovea there are only (rods/cones).
 - c) The peripheral retina contains (more rods than cones/only rods).
 - d) In contrast to the rods, the cones adapt (quicker/slower) to darkness, but (do/do not) reach a high level of light sensitivity.

Lecture 1: Basics of vision I - the eye

- e) Visual acuity is higher during (rod/cone) vision.
- f) Cone vision is (photopic/scotopic) vision which (is/is not) colour sensitive.
- g) Rod vision is (day/night) vision which operates best at (low/high) luminance.
- h) The Purkinje shift describes the increased sensitivity to (short/long) wavelengths in the (dark-/light-)adapted eye when vision is mainly (rod-/cone-)based.

Lecture 2: Basics of vision II - the brain

- **In which order does visual information run through the retino-geniculo-striate pathway?**

Pick the correct answer.

- a) Retina, lateral geniculate nucleus, optic chiasm
- b) Optic tract, lateral geniculate nucleus, optic radiations
- d) Optic nerve, optic radiations, optic tract
- c) Optic chiasm, optic nerve, V1

Lecture 2: Basics of vision II - the brain

- **Ungerleider & Mishkin (1982)**
describe two cortical visual
pathways.
Combine for both of them.

The (1-4) stream runs (a-d) and processes (i-iv) objects are.

1) frontal

a) parietal

i) why

2) ventral

b) central

ii) where

3) dorsal

c) temporal

iii) what

4) medial

d) lateral

iv) when

Lecture 3: The psychology of perception

- **Sort the statements: Who (1-5) said what (a-e)?**

1) Structuralism

a) Perception is the sum of measurable sensations.

2) Psychophysicists

b) Perception is top-down guided.

3) Ecologists (Gibson)

c) Perception is what matches best the good Gestalt.

4) Constructivists (Gregory)

d) Perception is the sum of sensory elements.

5) Gestalt psychologists

e) Perception is direct sensation and free of error.

Lecture 3: The psychology of perception

- **Psychophysics is all about perception thresholds.**

Do you remember the methods used to measure absolute thresholds (minimum intensity at which a stimulus is detected)?

Which of them returns the ...

1) ...self-controlled intensity to detect a stimulus?

Lecture 3: The psychology of perception

- **Psychophysics is all about perception thresholds.**

Do you remember the methods used to measure absolute thresholds (minimum intensity at which a stimulus is detected)?

Which of them returns the ...

2) ...mean threshold with which a stimulus is detected?

Lecture 3: The psychology of perception

- **Psychophysics is all about perception thresholds.**

Do you remember the methods used to measure absolute thresholds (minimum intensity at which a stimulus is detected)?

Which of them returns the ...

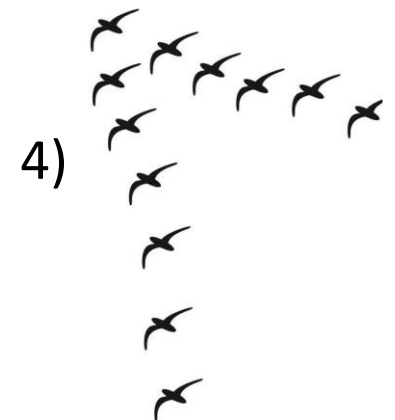
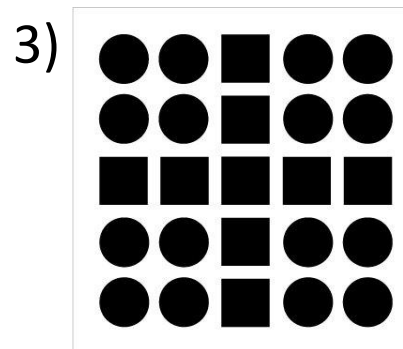
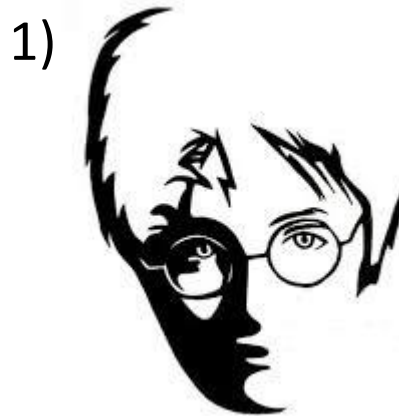
3) ...probability with which a stimulus is detected?

Lecture 4: When perception goes wrong

- **Gestalt psychologists suggest that we perceive objects in that we interpret retinal images according to a number of principles.**

Which principles (a-i) are illustrated here (1-4)?

- a) Similarity
- b) Proximity
- c) Common fate
- d) Continuation
- e) Closure
- f) Relative size
- g) Surroundedness
- h) Orientation
- i) Symmetry



Lecture 4: When perception goes wrong

- In this image one can either see 5 people or 6 pillars.

Why is this? Select from a-d:

What would Gregory say?

What would Gibson say?



- a) When sensation is unchanged, we perceive the same thing.
- b) When sensation changes, we perceive different things.
- c) Sensation is a matter of interpretation.
- d) We can trick perception when we carefully construct visual stimulation.

Lecture 5: Colour perception

- **Which alternative leads to a correct sentence?**
 - a) A tomato selectively (reflects/transmits) (long/medium/ short) wavelengths.
 - b) Milk (absorbs/transmits) (some/all) wavelengths.
 - c) How to describe a colour:
Colour values are specified in (pixels/hues),
colour saturation specifies
(intensity of light/the amount of white added),
and brightness the amount of
(white added/light reflected).

Lecture 5: Colour perception

- **Which alternative leads to a correct sentence?**

d) The trichromatic theory suggests that any colour can be matched by correctly adjusting the proportions of (two/three/four) wavelengths.

e) It takes at least (two/three/four) functioning types of cones for full colour perception.

f) The opponent-process theory suggests that no colour can be described as a (reddish/yellowish/blueish) green.

g) Trichromatic and opponent colours (cannot coexist/run in parallel/are processed sequentially).

Lecture 6: Depth perception

- Find an example for each of the 8 pictorial depth cues.



1) Familiar size

2) Relative height

3) Shadows

4) Atmospheric perspective

5) Relative size

6) Texture gradient

7) Perspective convergence

8) Occlusion

Lecture 6: Depth perception

- **For each statement, decide whether it's right or wrong?**
 - a) Stereopsis is the ability to use retinal disparity as a cue to perceive depth.
 - b) Stereopsis is the ability to use binocular disparity as a cue to perceive depth.
 - c) Retinal disparity is not completely sufficient to produce stereopsis – additional cues are needed.
 - d) Retinal disparity describes a shift of retinal images in our two eyes.

Lecture 6: Depth perception

- **For each statement, decide whether it's right or wrong?**
 - e) The horopter is an imaginary plane connecting all objects that are equally far away from the observer than the fixation point.
 - f) All objects on the horopter produce corresponding retinal images, i.e., the locations of the two retinal images are mirror-reversed in both eyes.

Lecture 6: Depth perception

- **For each statement, decide whether it's right or wrong?**
 - g) With respect to the fixation point, each distance in the visual field produces a different amount of retinal disparity.
 - h) The larger the disparity the closer to fixation an object is.

Lecture 6: Size perception

- **How does this visual illusion work?**



The visual angle of the can is (smaller/larger) than the visual angle of the person, and because the perceived distance of the can and the person is (the same/different), we perceive an oversized can.

Lecture 7: Motion perception

- **For each statement, decide whether it's right or wrong?**
 - a) Almost all neurons in MT are motion-selective.
 - b) Single MT neurons can reliably detect motion direction.
 - c) Biological motion produces retinal motion.
 - d) A static object in a moving eye leads to motion perception because there is an image displacement signal.

Lecture 7: Motion perception

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 - c) Biological motion produces retinal motion.
 - d) A static object in a moving eye leads to motion perception because there is an image displacement signal.
 - e) A moving object, tracked by a moving eye, elicits motion perception as there are image displacement and motor signals.

Lecture 8: Object perception and recognition

- **David Marr (1982) proposed 4 levels of visual processing.**
Combine from the table: What (a-d) happens when (1-4)?

- | | |
|-----------------------|--|
| 1. Raw primal sketch | a. Computation of a standard form. |
| 2. Full primal sketch | b. Computation of edges and contours. |
| 3. 2½-D sketch | c. Computation of light intensity changes. |
| 4. 3-D sketch | d. Computation of locations and depth. |

Lecture 8: Object perception and recognition

- **Which answer is wrong?**

People with apperceptive agnosia cannot...

- a) ... perceive objects.
- b) ... recognise objects.
- c) ... copy shapes.
- d) ... see objects.

Lecture 8: Object perception and recognition

- **Which answer is wrong?**

People with associative agnosia cannot...

- a) ... retrieve meaning of objects.
- b) ... create viewer-centred representations.
- c) ... name objects.
- d) ... recognise objects.

Good luck with your exams!!



You got this!